

Statistics Basics| Assignment

Q 1 What is the difference between descriptive statistics and inferential statistics? Explain with examples.

Ans - The difference between **descriptive statistics** and **inferential statistics** is mainly about what they do with data:

1. Descriptive Statistics

Meaning:

Descriptive statistics are used to **summarize and describe the data you already have**.

They do not go beyond the data—no predictions or generalizations.

Common tools:

- Mean (average)
- Median
- Mode
- Range
- Standard deviation
- Tables, graphs, charts

Example:

Suppose a teacher has marks of 5 students:

60, 70, 80, 90, 100

2. Inferential Statistics

Meaning:

Inferential statistics are used to **make conclusions or predictions about a larger population based on a sample of data**.

They involve estimation, hypothesis testing, and generalization.

Common tools:

- Hypothesis testing (t-test, chi-square test)
- Confidence intervals
- Regression analysis

Q 2 What is sampling statistics? Explain the differences between random and stratified sampling.

Ans - Sampling in Statistics

Sampling is the process of selecting a **small group (called a sample)** from a **larger group (called a population)** to study and draw conclusions about the whole population.

Random Sampling

Meaning:

In **random sampling**, every individual in the population has an **equal chance of being selected**.

Selection is purely by chance.

Stratified Sampling

Meaning:

In **stratified sampling**, the population is first divided into **subgroups (called strata)** based on a characteristic (like age, gender, class, income), and then samples are taken from **each subgroup**.

Q 3 Define mean, median, and mode. Explain why these measures of central tendency are important.

Ans - Measures of Central Tendency: Mean, Median, and Mode

. Mean

Definition:

The **mean** is the **average** of all values in a dataset. It is calculated by adding all values and dividing by the number of values.

Formula:

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}} \quad \text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Example:

Data: 10, 20, 30

$$\text{Mean} = (10 + 20 + 30) \div 3 = 60 \div 3 = \mathbf{20}$$

2. Median

Definition:

The **median** is the **middle value** when data is arranged in ascending or descending order.

Example:

Data: 5, 10, 15

Median = **10**

If the number of values is even:

Data: 5, 10, 15, 20

Median = $(10 + 15) \div 2 = \mathbf{12.5}$

3. Mode

Definition:

The **mode** is the value that appears **most frequently** in a dataset.

Example:

Data: 2, 3, 3, 5, 7

Mode = **3** (because it appears most often)

Q 4 Explain skewness and kurtosis. What does a positive skew imply about the data?

Ans - Skewness

Skewness measures the **asymmetry** of a data distribution—i.e., whether the data is evenly distributed or tilted toward one side.

Types of skewness:

1. **Symmetrical distribution (zero skewness)**
 - a. Left and right sides are balanced
 - b. Mean = Median = Mode
2. **Positive skew (right-skewed distribution)**
 - a. Tail extends more to the **right (higher values)**
 - b. Most data values are concentrated on the left
3. **Negative skew (left-skewed distribution)**
 - a. Tail extends more to the **left (lower values)**
 - b. Most data values are concentrated on the right

What does **positive skewness** imply?

A **positive skew** means:

- There are a few **very high values (outliers)** pulling the distribution to the right
- Most data values are **small or moderate**

- Typically:

$Mean > Median > Mode$ | $text{Mean} > |text{Median} > |text{Mode}$ Mean > Median > Mode

Example:

Income distribution in a country

- Most people earn moderate income
- A few people earn extremely high incomes → pulls the average upward'

Q 5 Implement a Python program to compute the mean, median, and mode of a given list of numbers. numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28] (Include your Python code and output in the code box below.)

Ans -

Python program to compute mean, median, and mode

```
import statistics
numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]
```

Mean

```
mean_value = statistics.mean(numbers)
```

Median

```
median_value = statistics.median(numbers)
```

Mode

```
mode_value = statistics.mode(numbers)
print("Numbers:", numbers) print("Mean:", mean_value) print("Median:", median_value)
print("Mode:", mode_value)
```

Python program to compute mean, median, and mode

```
import statistics
numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]
```

Mean

```
mean_value = statistics.mean(numbers)
```

Median

```
median_value = statistics.median(numbers)
```

Mode

```
mode_value = statistics.mode(numbers)
print("Numbers:", numbers) print("Mean:", mean_value) print("Median:", median_value)
print("Mode:", mode_value)
```

Q 6 Compute the covariance and correlation coefficient between the following two datasets provided as lists in Python: list_x = [10, 20, 30, 40, 50] list_y = [15, 25, 35, 45, 60] (Include your Python code and output in the code box below.)

Ans -

Compute covariance and correlation coefficient

```
import numpy as np
list_x = [10, 20, 30, 40, 50] list_y = [15, 25, 35, 45, 60]
x = np.array(list_x) y = np.array(list_y)
```

Mean

```
mean_x = np.mean(x) mean_y = np.mean(y)
```

Covariance

```
covariance = np.mean((x - mean_x) * (y - mean_y))
```

Standard deviations

```
std_x = np.std(x) std_y = np.std(y)
```

Correlation coefficient

```
correlation = covariance / (std_x * std_y)
print("List X:", list_x) print("List Y:", list_y) print("Covariance:", covariance) print("Correlation Coefficient:", correlation)
```

Q 7 Write a Python script to draw a boxplot for the following numeric list and identify its outliers.

Explain the result: data = [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]

Ans -

Boxplot and outlier detection

```
import matplotlib.pyplot as plt
import numpy as np
data = [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]
```

Convert to numpy array

```
x = np.array(data)
```

Quartiles

```
Q1 = np.percentile(x, 25)
Q3 = np.percentile(x, 75)
IQR = Q3 - Q1
```

Outlier boundaries

```
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
```

Detect outliers

```
outliers = x[(x < lower_bound) | (x > upper_bound)]
print("Data:", data)
print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)
print("Outliers:", outliers)
```

Boxplot

```
plt.boxplot(x)
plt.title("Boxplot of Given Data")
plt.ylabel("Values")
plt.show()
```

Q 8 You are working as a data analyst in an e-commerce company. The marketing team wants to know if there is a relationship between advertising spend and daily sales. ● Explain how you would use covariance and correlation to explore this relationship. ● Write Python code to compute the correlation between the two lists: advertising_spend = [200, 250, 300, 400, 500] daily_sales = [2200, 2450, 2750, 3200, 4000]

Ans - . Using Covariance and Correlation to Study the Relationship

As a data analyst, you would use **covariance** and **correlation** to understand whether **advertising spend affects daily sales** and how strongly they are related.

Covariance (Direction of relationship)

- Covariance tells whether two variables move **together or in opposite directions**.
- If covariance is:

- **Positive** → when advertising spend increases, sales also increase
- **Negative** → when advertising increases, sales decrease
- **Zero** → no relationship

Limitation: Covariance does **not tell the strength** of the relationship.

```
import numpy as np

advertising_spend = [200, 250, 300, 400, 500]
daily_sales = [2200, 2450, 2750, 3200, 4000]

x = np.array(advertising_spend)
y = np.array(daily_sales)

# Mean
mean_x = np.mean(x)
mean_y = np.mean(y)

# Covariance
covariance = np.mean((x - mean_x) * (y - mean_y))

# Standard deviations
std_x = np.std(x)
std_y = np.std(y)

# Correlation coefficient
correlation = covariance / (std_x * std_y)

print("Advertising Spend:", advertising_spend)
print("Daily Sales:", daily_sales)
print("Covariance:", covariance)
print("Correlation Coefficient:", correlation)
```

Q 9 Your team has collected customer satisfaction survey data on a scale of 1-10 and wants to understand its distribution before launching a new product. ● Explain which summary statistics and visualizations (e.g. mean, standard deviation, histogram) you'd use. ● Write Python code to create a histogram using Matplotlib for the survey data: survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10, 4, 7, 6, 9, 8, 7]

Ans - **summary statistics** and **visual tools**.

Summary Statistics

These help describe the dataset in numbers:

- **Mean (Average):**
Shows overall satisfaction level. A higher mean means customers are generally satisfied.
- **Median (Middle value):**
Useful if there are extreme scores (very low or very high ratings). It shows the “typical” response.
- **Mode (Most frequent value):**
Shows the most common satisfaction score (e.g., most customers may rate 7 or 8).
- **Standard Deviation:**
Measures how spread out the responses are.
 - Low value → customers feel similarly
 - High value → mixed opinions

Visualizations

These help understand the shape of the data:

- **Histogram:**
 - Shows how often each score appears
 - Helps identify whether ratings are skewed (e.g., mostly high or low)
- **Boxplot (optional):**
 - Shows median, quartiles, and outliers
 - Helps detect unusual ratings